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Introduction to Cloud Computing

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- What is Cloud Computing ?
- Cloud Computing Characteristics
- Cloud service models
- Cloud deployment models
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The Cloud Computing Phenomenon

- **Significant Adoption Across Organizations**
 - Cloud computing adoption is rising rapidly across industries.
 - Organizations of all sizes leverage cloud technologies for efficiency, scalability, and cost-efficient.
- **A Leading "Disruptive" Technology**
 - Recognized as a game-changer in the IT landscape.
 - Transforms traditional IT infrastructure, making it more flexible and accessible.
- **Driving Business Model Optimization and Innovation**
 - Enables businesses to optimize operations, efficient cost, and improve agility.
 - Fosters innovation by providing tools for AI, machine learning, big data, and IoT.
 - Supports the creation of new revenue streams and digital-first business models.

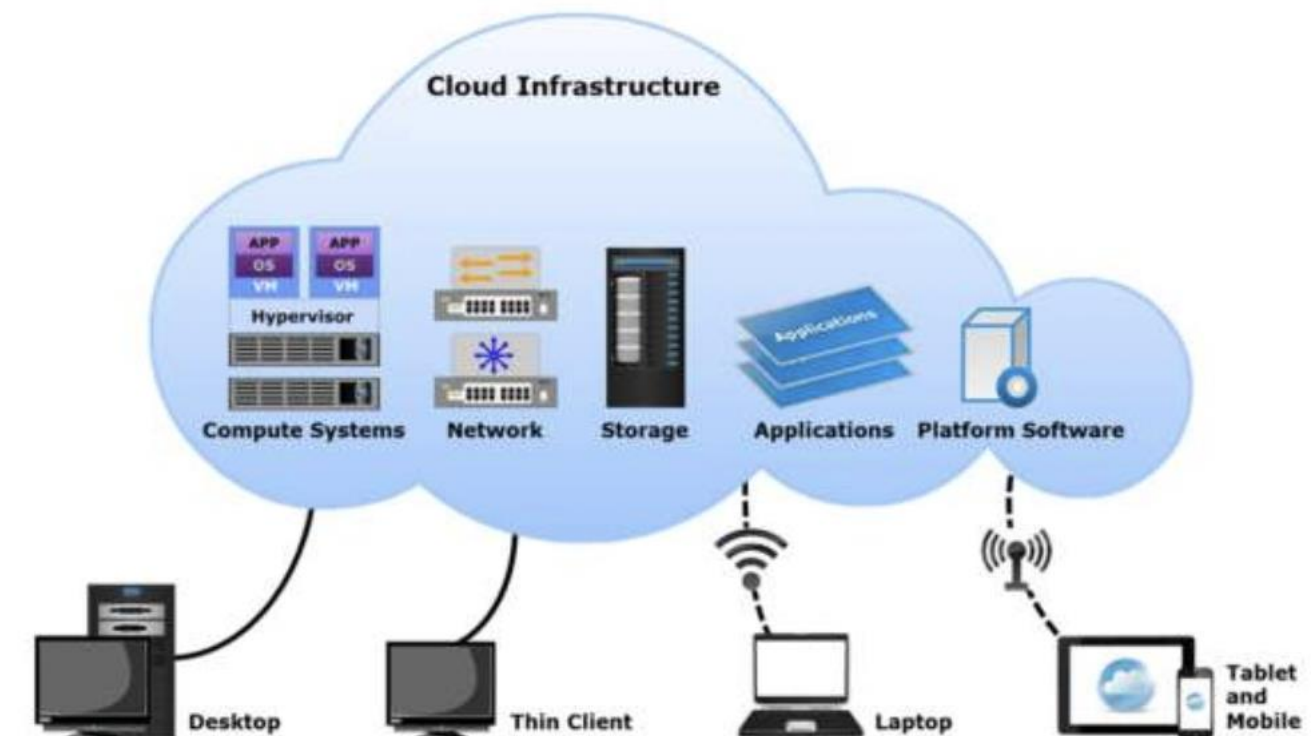


What is cloud computing ?

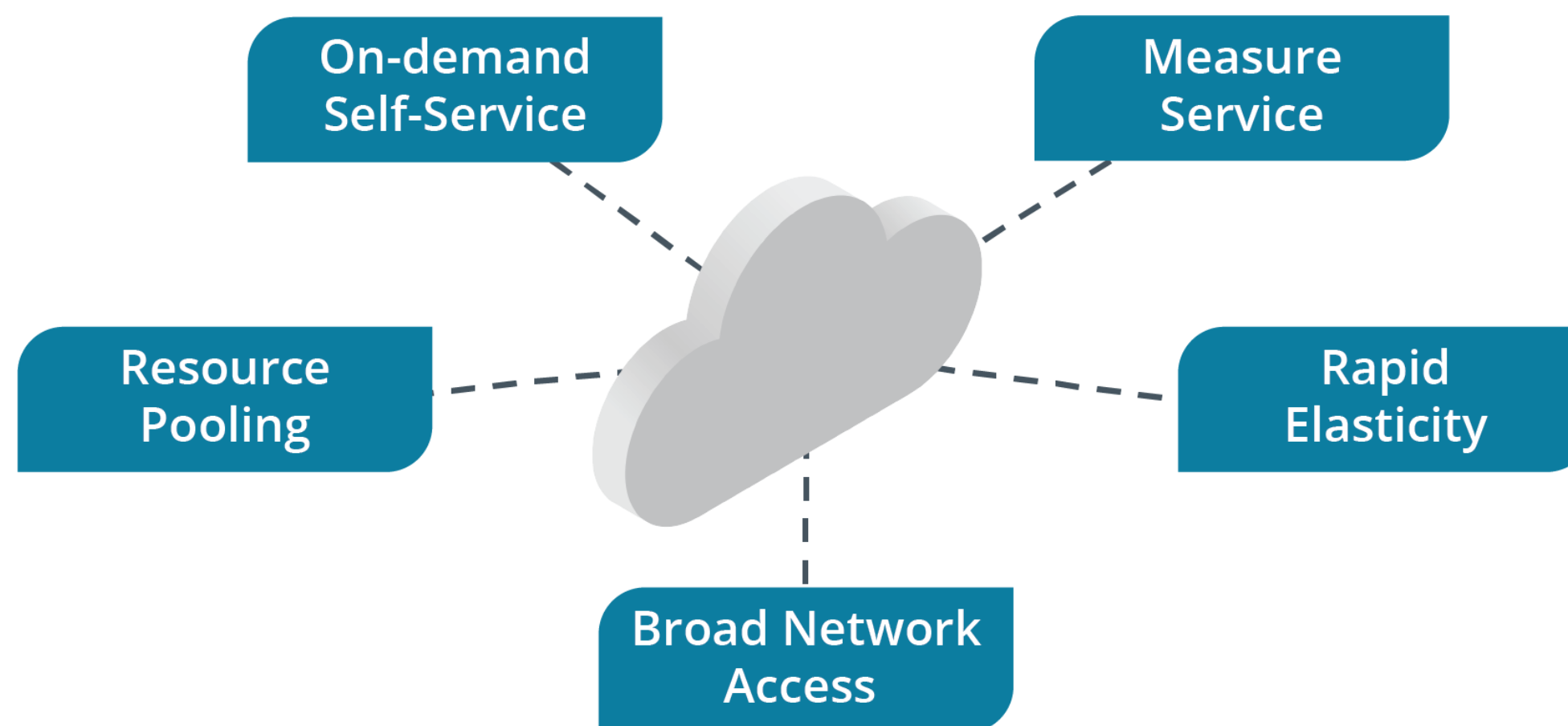
Cloud Computing

A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources, (e.g., servers, storage, networks, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

– U.S. National Institute of Standards and Technology, Special Publication 800-145



Cloud Characteristics



On-demand Self-service

- Services are consumed on an as-needed basis without the need for human assistance from the Cloud Service Provider.
- Organizations provision resources such as servers, storage, or web applications when needed.
- Organizations can provision resources by using their own internal IT staff without relying on the Cloud Service Provider's staff.



Broad Network Access

Computing capabilities are available over the network

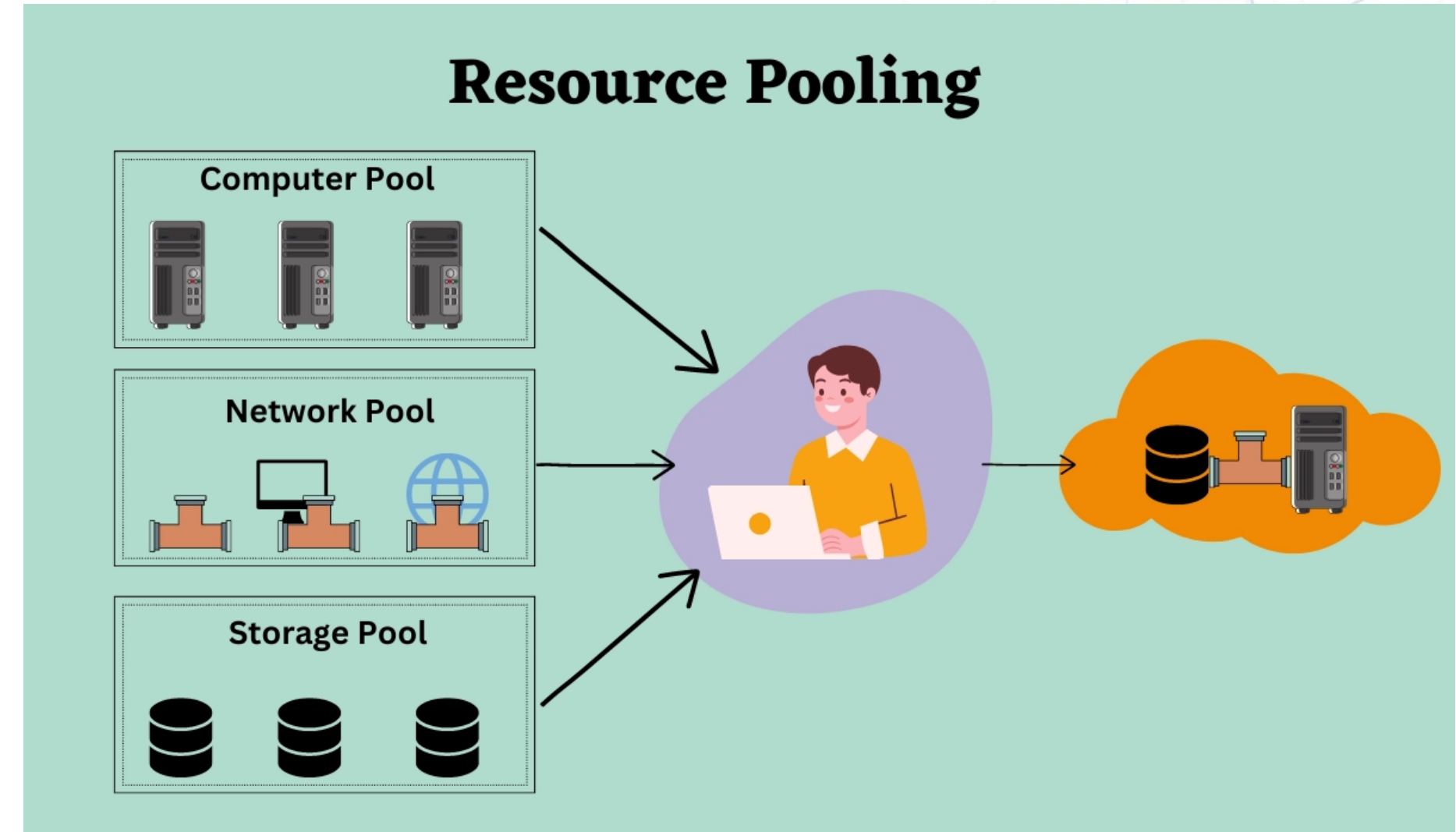
Computing capabilities are accessed from a broad range of client platforms such as:

- ▶ Desktop computer
- ▶ Laptop
- ▶ Tablet
- ▶ Mobile device



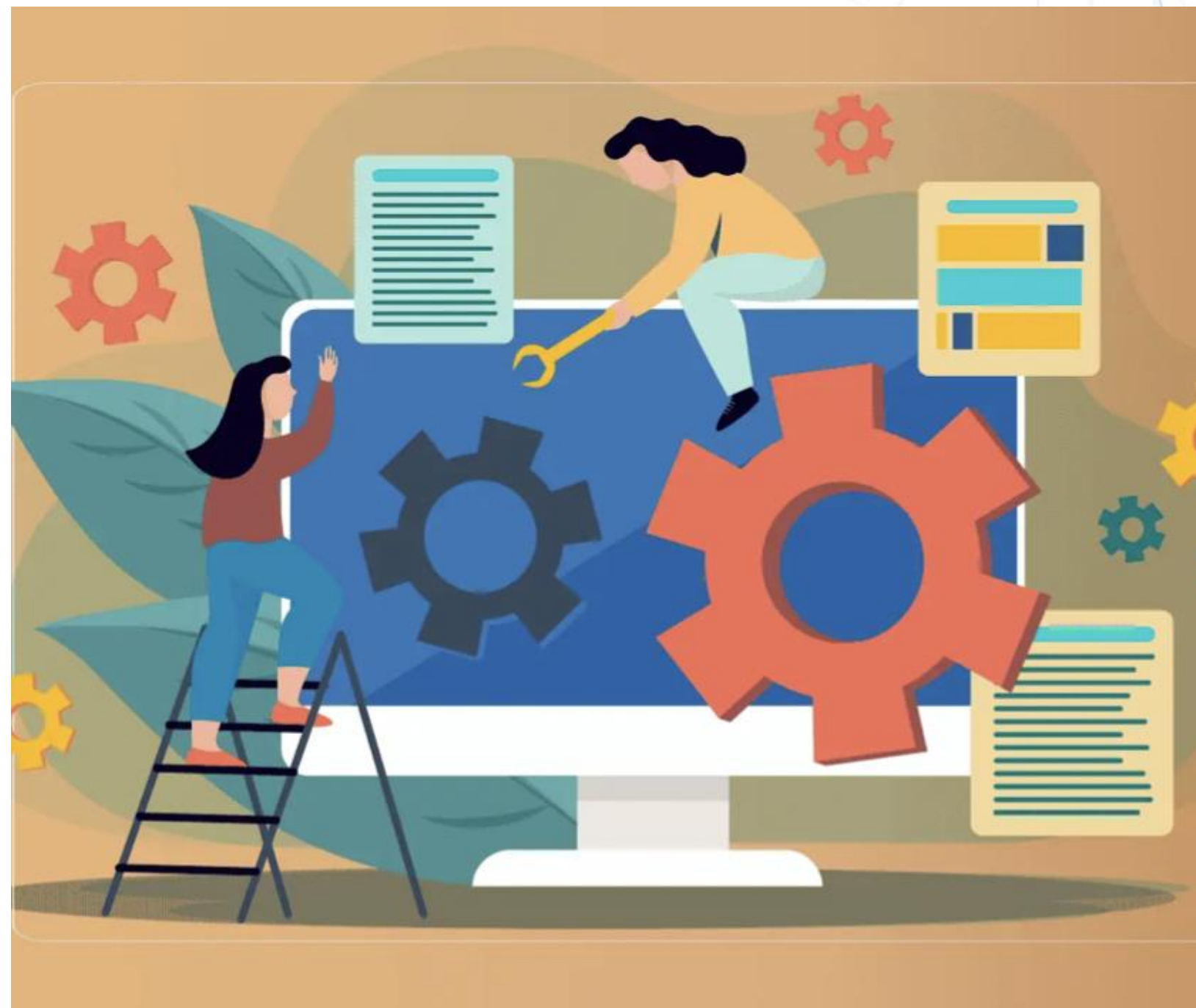
Resource Pooling

- Cloud service providers:
 - Pool network, storage, and compute capabilities.
 - Allocate these pooled resources to customers on demand.
 - Take advantage of the economy of scale.



Rapid Elasticity

- Scale computing resources—such as storage, processing power, or memory—up or down quickly and efficiently, based on the current demand.

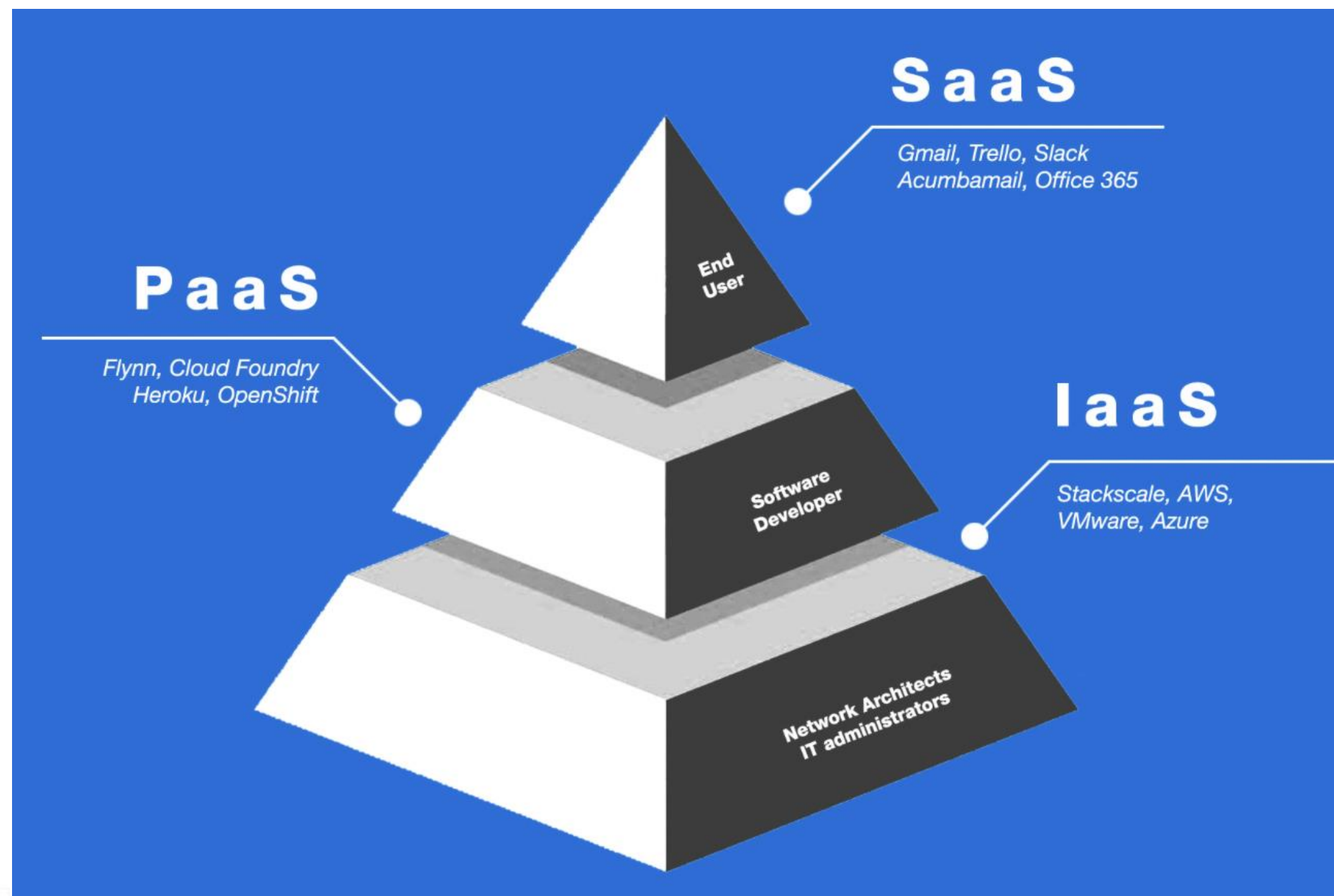


Measured Services

- Cloud resource utilization is carefully metered by cloud service providers.
- Billing for cloud resources is based on actual use of the resources.



Cloud Service models



SaaS

On-site	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

-  You manage
-  Service provider manages

PaaS

On-site	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

-  You manage
-  Service provider manages

IaaS

On-site	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

-  You manage
-  Service provider manages

XaaS

Catch-all phrase for any technology solutions that are moved to the cloud or that exhibit the same pay-as-you-go subscription model as the three primary cloud service models.

Everything as a service (XaaS)



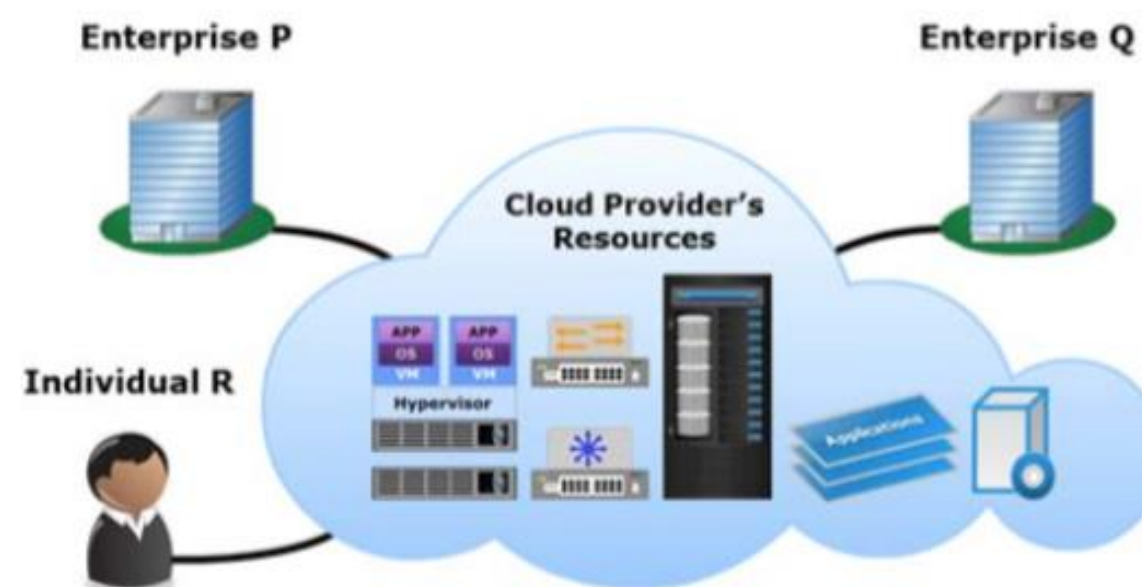
Cloud deployment models

Public Cloud

Public Cloud

The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.

– U.S. National Institute of Standards and Technology, Special Publication 800-145



Private Cloud

Private Cloud

Private Cloud

The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (for example, business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.

– U.S. National Institute of Standards and Technology, Special Publication 800-145

- There are two variants of private cloud:
 - On-premise
 - Externally-hosted

Community Cloud

Community Cloud

Community Cloud

The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

– U.S. National Institute of Standards and Technology, Special Publication 800-145

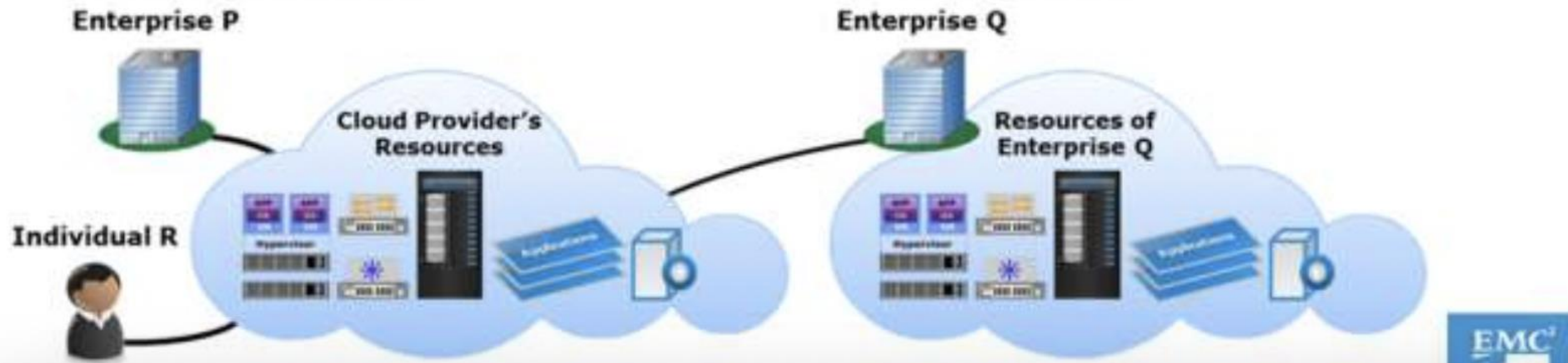
- There are two variants of community cloud:
 - On-premise
 - Externally-hosted

Hybrid Cloud

Hybrid Cloud

The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

– U.S. National Institute of Standards and Technology, Special Publication 800-145



The evolution of cloud computing

1. Pre-Cloud Era (1950s-1990s)

1950s-1970s: Foundation Era

- **Mainframe Computing (1950s-1960s)**
 - Large, centralized computers in dedicated data centers
 - IBM dominated the mainframe market
- **Time-Sharing Systems (1960s-1970s)**
 - Multiple users could access mainframe resources simultaneously
 - UNIX development began at Bell Labs (1969)

1980s-1990s: Distributed Computing

- **Client-Server Architecture (1980s)**
 - Personal computers became widespread
 - Local Area Networks (LANs) enabled resource sharing
 - Distributed computing concepts emerged
- **Early Virtualization (1990s)**
 - VMware founded (1998)
 - First x86 virtualization products released (1999)
- **Internet Service Providers (1990s)**
 - Web hosting services emerged
 - Early Application Service Providers (ASPs) appeared
 - Telecommunications companies began offering Virtual Private Networks (VPNs)



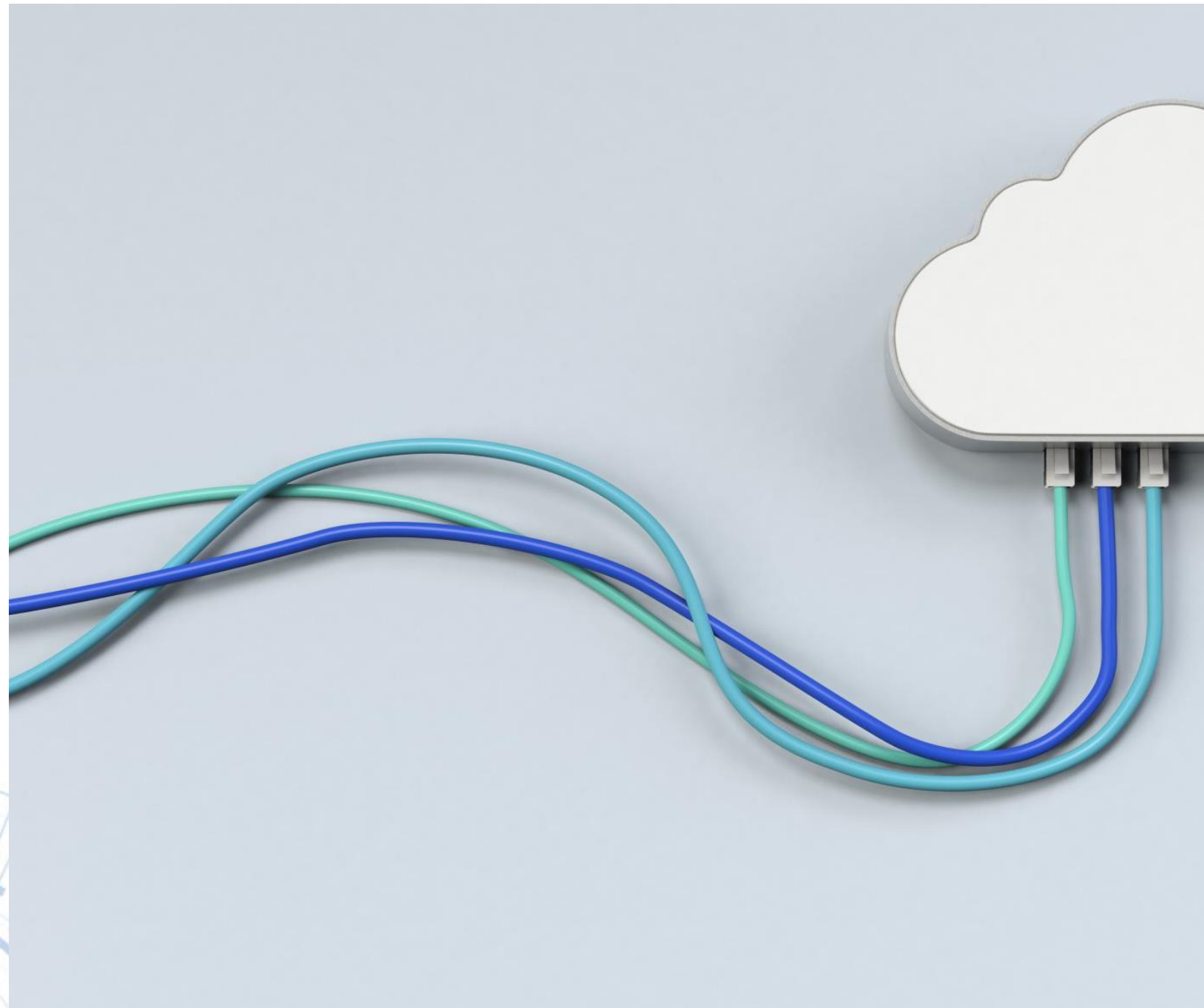
The evolution of cloud computing

2. Birth of Modern Cloud Computing (1999-2006)

- Early Innovations
- 1999:
 - Salesforce launches as one of the first successful SaaS companies
 - VMware introduces virtual machine technology for x86 systems
- 2000-2001:
 - Early grid computing projects gain traction
 - Concept of utility computing resurfaces
- 2002:
 - Amazon Web Services launches with initial web services
 - First basic storage and computation services offered
- 2006:
 - Amazon EC2 (Elastic Compute Cloud) launches
 - Modern cloud computing era officially begins
 - Pay-as-you-go compute resources become available



The evolution of cloud computing



3. Cloud Computing Growth (2007-2014)

- **Major Platform Launches**

- **2007-2008:**

- Google Docs becomes free to all users
- Google App Engine launches (2008)
- Eucalyptus, first open-source AWS-compatible platform, releases

- **2009-2010:**

- Microsoft Azure launches (initially as Windows Azure)
- OpenStack project begins
- Rackspace launches Cloud Servers product

- **2011-2012:**

- Google Cloud Platform launches
- IBM SmartCloud Enterprise debuts
- Oracle Cloud launches

- **Technology Advances**

- **2013-2014:**

- Docker containerization technology released (2013)
- AWS Lambda introduces serverless computing (2014)
- Major providers begin rapid service expansion
- Cloud storage prices begin dramatic decline

The evolution of cloud computing



4. Maturation Era (2015-2019)

- **Cloud-Native Technologies**
- **Container Orchestration**
 - Kubernetes released by Google (2015)
 - Container adoption accelerates
 - Microservices architecture becomes mainstream
- **DevOps Integration**
 - CI/CD pipelines become standard
 - Infrastructure as Code (IaC) gains prominence
 - Automation tools proliferate
- **Industry Consolidation**
- Market leaders emerge clearly:
 - AWS dominates market share
 - Microsoft Azure establishes strong enterprise presence
 - Google Cloud Platform gains traction
- Enterprise adoption accelerates
- Hybrid cloud models mature

The evolution of cloud computing



Modern Era (2020-Present)

- **Current Trends**
- **Multi-Cloud Strategy**
 - Organizations leverage multiple cloud providers
 - Cloud-agnostic tools become essential
 - Focus on avoiding vendor lock-in
- **Edge Computing**
 - Processing moves closer to data sources
 - 5G enables new edge use cases
 - IoT drives edge computing adoption
- **AI/ML Integration**
 - Cloud providers offer comprehensive AI services
 - Machine learning becomes more accessible
 - AI-optimized infrastructure emerges

Key benefits of Cloud Computing

C Cost Benefits

- No upfront infrastructure costs
- Pay-as-you-go pricing model
- Reduced maintenance costs

T Technical Advantages

- Automatic scaling based on demand
- High-performance computing capabilities
- Built-in redundancy and reliability

B Business Agility

- Rapid resource provisioning
- Faster time to market
- Global market reach

E Enhanced Security

- Enterprise-grade security features
- Regular security updates
- Advanced threat protection

Key Benefits of Cloud Computing

D Data Analytics

- Advanced analytics capabilities
- Real-time data processing
- Machine learning integration

Claude content

S Sustainability

- Reduced carbon footprint
- Optimized energy consumption
- Green data center initiatives

O Operational Benefits

- Simplified management
- Improved collaboration
- Focus on core business

B Business Continuity

- Automated backup systems
- Geographic redundancy
- 24/7 system availability

Key Challenges in Cloud Computing

S Security and Privacy

- Data breaches and unauthorized access risks
- Compliance with regulations (GDPR, HIPAA)
- Multi-tenancy security concerns

T Technical Issues

- Network latency and performance inconsistency
- Service outages and reliability concerns
- Legacy system integration complexity

C Cost Management

- Unexpected costs and billing surprises
- Resource over-provisioning waste
- Hidden costs in data transfer and API calls

V Vendor Dependencies

- Vendor lock-in problems
- Limited customization options
- Service provider reliability concerns

Key Challenges in Cloud Computing

D Data Management

- Complex data migration processes
- Data governance and consistency issues
- Backup and recovery challenges

O Operational Issues

- Skilled personnel shortage
- Continuous training requirements
- Change management difficulties

O Operational Issues

- Skilled personnel shortage
- Continuous training requirements
- Change management difficulties

N Network Concerns

- Internet dependency risks
- Bandwidth limitations
- Geographic distribution challenges

AWS

- AWS: Leading cloud service platform offering scalable, secure, and cost-efficient solutions globally.
- Launched in 2006 with S3 and EC2.
- Global presence: Operates in multiple regions with high availability and compliance.
- Revenue in 2024: \$110 billion annual run rate.
- Over 200 fully featured services across computing, storage, databases, machine learning, and more.
- Millions of active customers, including startups, enterprises, and governments.

Value Proposition:

- Pay-as-you-go pricing model.
- Broad service portfolio tailored for industries like healthcare, finance, and retail.
- Strong security and compliance with global standards.

Top Companies using Amazon Web Services (AWS)

- *Netflix*
- *Spotify*
- *Airbnb*
- *Uber*
- *Peloton*
- *Expedia*
- *Pinterest*
- *Samsung*
- *Sony*
- *Novartis*

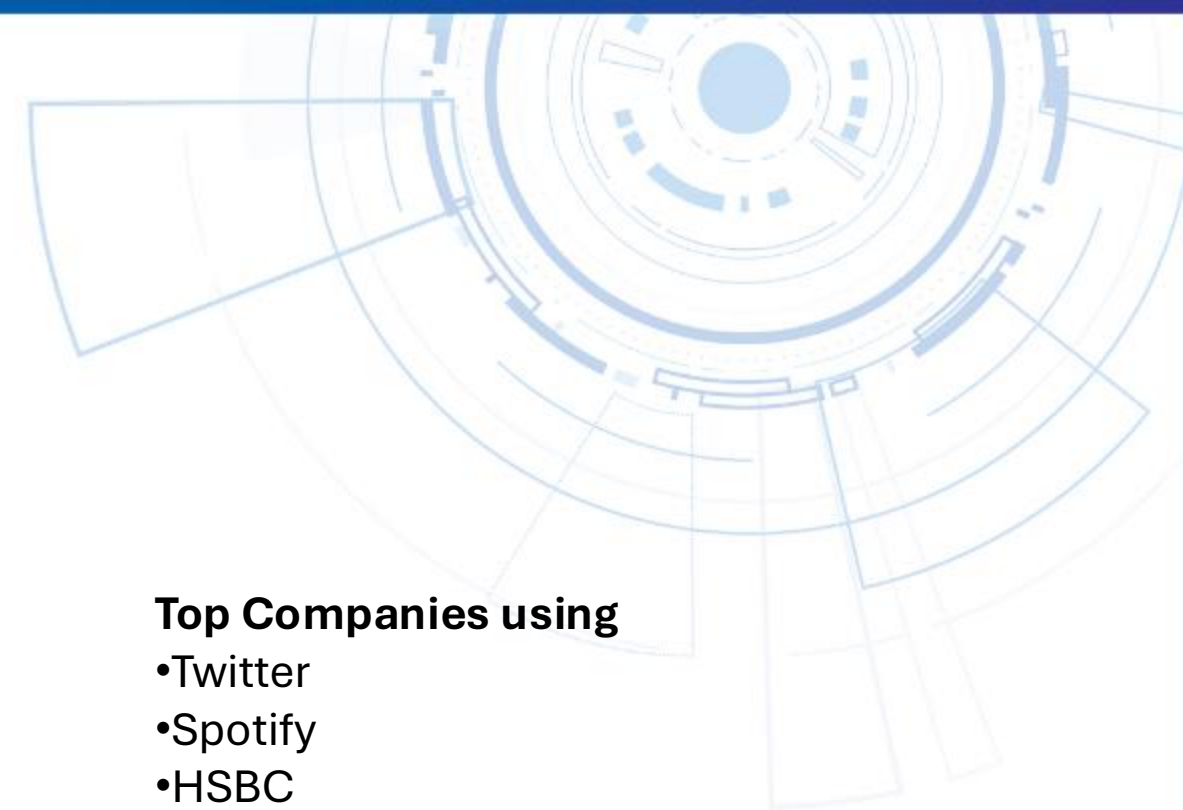
Azure

- Launched in **2010** as *Windows Azure*, renamed **Microsoft Azure** in **2014**
- One of the **fastest-growing cloud platforms**, competing directly with **AWS**
- Global infrastructure: **54+ regions**, available in **140+ countries**
- Comprehensive services: **Compute, Storage, Databases, Networking, Security, AI & Machine Learning, Analytics, DevOps, Containers, IoT, Identity**
- Supports **public, private, and hybrid cloud** deployments
- Strong integration with **Microsoft ecosystem** (Windows, Active Directory, Office)
- High **scalability, security, flexibility, and cost-effectiveness**
- **30-day free trial** with enterprise-grade support

Top Companies using Microsoft Azure

- *Walmart*
- *Macy's*
- *The Home Depot*
- *Starbucks*
- *Coca-Cola*
- *Bank of America*
- *JPMorgan Chase*
- *Citigroup*
- *Fidelity Investments*
- *Standard Chartered*

Google Cloud Platform (GCP)



3. Google Cloud Platform

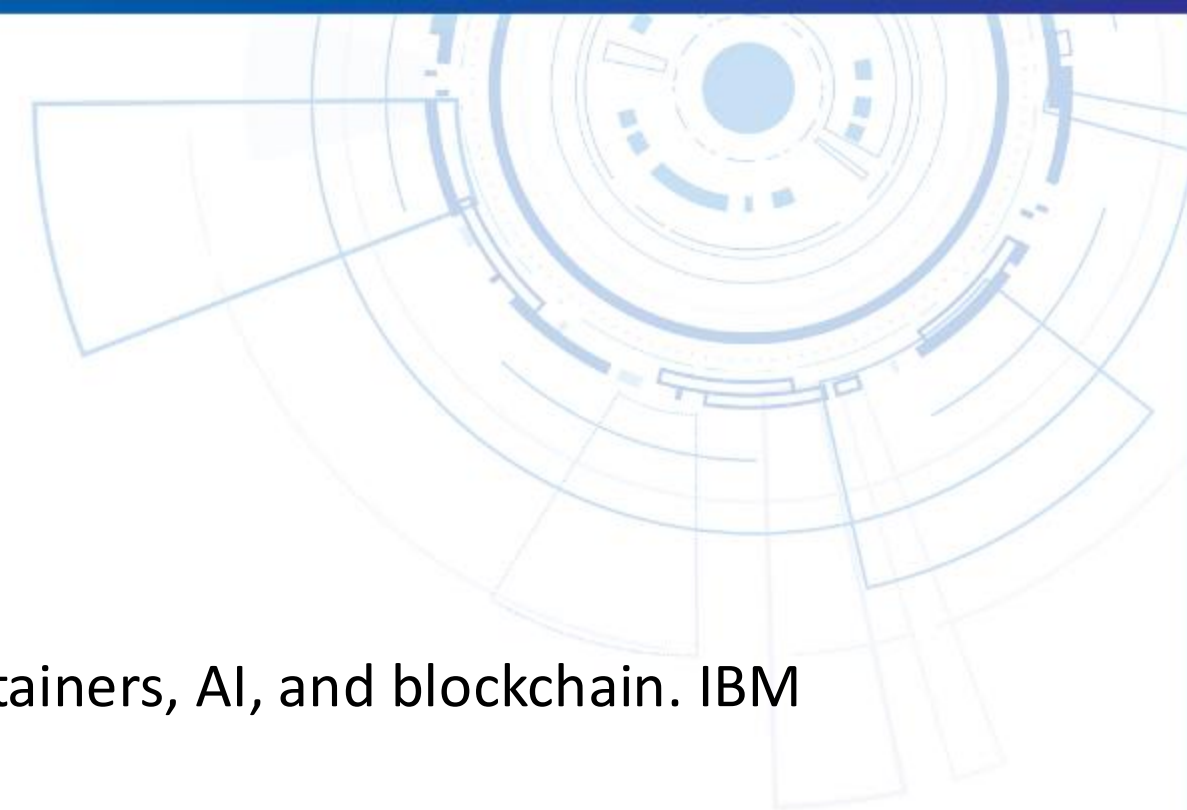
- Launched in **2008** with **App Engine**, entering the **PaaS** market
- Evolved into a **full-scale cloud provider**, competing with **AWS** and **Microsoft Azure**
- Strong leadership in **Big Data, Analytics & Machine Learning**
- Key services: **BigQuery, Vertex AI**
- Global infrastructure: **38 regions**, serving **200+ countries**
- Commitment to **sustainability**: carbon-neutral & renewable energy data centers
- Broad service portfolio: **Compute, Storage, Databases, Networking, AI/ML, Big Data, GKE**
- Seamless integration with **Google Workspace** and **Android ecosystem**
- **90-day free trial** and **generous free tier** for new users

Top Companies using

- Twitter
- Spotify
- HSBC
- PayPal
- Snap Inc.
- Etsy
- Target
- ShareChat
- L'Oréal
- AccuWeather



Other Cloud Providers



- **IBM Cloud**

Provides a full stack cloud platform with over 170 products and services covering data, containers, AI, and blockchain. IBM Cloud is noted for its enterprise-grade solutions and strong emphasis on security.

- **Oracle Cloud**

Offers integrated cloud applications and platform services, including database services, enterprise applications, and infrastructure solutions. Oracle Cloud is known for its robust database services and enterprise solutions

- **Alibaba Cloud**

Provides a comprehensive suite of global cloud computing services, including elastic computing, data storage, and big data processing. Alibaba Cloud is a leading cloud provider in the Asia-Pacific region

- **Salesforce**

Specializes in customer relationship management (CRM) software and applications focused on sales and customer service. Salesforce is renowned for its CRM solutions and cloud-based applications.

- **RackSpace Technology**

Offers multi-cloud solutions, including cloud migration, management, and optimization services. Rackspace is known for its managed cloud services and support across various cloud platforms.

- **VMware Cloud**

Provides cloud infrastructure and management services, enabling businesses to run, manage, and secure applications across clouds and devices. VMware Cloud is recognized for its virtualization technologies and hybrid cloud solutions.

Case studies of Cloud Computing

Netflix – Global Video Streaming Service

- **Challenge:** Scaling to support millions of users while maintaining high availability and performance.
- **Solution:**
 - Migrated to **Amazon Web Services (AWS)** to leverage scalability and redundancy.
 - Used **EC2 instances** for compute and **S3** for storage.
- **Outcome:**
 - Achieved seamless scaling during peak demand (e.g., new series releases).
 - Enhanced global content delivery using AWS's Content Delivery Network (CDN).

Case studies of Cloud Computing

Coca-Cola – Marketing and Analytics

- **Challenge:** Analyzing massive amounts of data for targeted marketing campaigns.
- **Solution:**
 - Deployed on **Google Cloud Platform (GCP)**.
 - Used **BigQuery** for data analytics and **Google Kubernetes Engine (GKE)** for scalable applications.
- **Outcome:**
 - Gained insights into customer behavior for personalized campaigns.
 - Reduced data processing time significantly.

Case studies of Cloud Computing

Zoom – Video Conferencing Platform

- **Challenge:** Scaling rapidly to meet unprecedented demand during the COVID-19 pandemic.
- **Solution:**
 - Partnered with **AWS** for elastic scaling of compute and network resources.
 - Used **Azure** for additional cloud capacity.
- **Outcome:**
 - Handled exponential growth in daily users (from 10M to 300M in 2020).
 - Maintained reliability and performance under pressure.

Case studies of Cloud Computing

General Electric (GE) – Industrial IoT

- **Challenge:** Managing and analyzing massive industrial data from IoT devices.
- **Solution:**
 - Built its **Predix** platform on **Microsoft Azure**.
 - Integrated Azure IoT services for real-time analytics.
- **Outcome:**
 - Improved operational efficiency by analyzing machine performance.
 - Enabled predictive maintenance, reducing downtime and costs.

Case studies of Cloud Computing

The Weather Company – Forecasting and Analytics

- **Challenge:** Processing vast amounts of weather data in real-time for accurate forecasting.
- **Solution:**
 - Migrated to **IBM Cloud**.
 - Used cloud-based analytics tools and IoT platforms.
- **Outcome:**
 - Delivered precise, real-time weather forecasts.
 - Enhanced decision-making for industries like aviation and logistics.